Homework — 1.2.4 Types of Programming Language

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.2.4 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Explain what is meant by a **programming paradigm**, and explain why no single paradigm is best for every problem. **[2]** *(AO1)*

2. State **three** features that characterise the **procedural** paradigm. **[3]** *(AO1)*

3. The following Little Man Computer (LMC) program reads two numbers and outputs a result.

```
         INP
         STA  first
         INP
         SUB  first
         BRP  positive
         OUT
         HLT
positive OUT
         HLT
first    DAT
```

(a) Describe in words what value this program outputs. **[2]** *(AO2)*

(b) State what the `BRP` instruction does and why it is used here. **[2]** *(AO1)*

4. Addressing-mode calculation. A processor executes `LDA 4`. The relevant memory contents are:

> Address 4 holds the value **7** · Address 7 holds the value **2** · The index register holds **3**.

State the value loaded into the Accumulator for each mode below. **[3]** *(AO2)*

(a) Immediate: __ **(b) Direct:** _ ***(c) Indirect:*** ___

(d) *Stretch — not marked:* State the *effective address* used in **indexed** addressing: _______

5. Explain the OOP concept of **inheritance** and give **one** benefit it provides to a programmer. **[2]** *(AO1)*

Part B — Retrieval: keep it fresh (~6 marks)

6. *(from 1.2.3)* State **two** advantages and **one** disadvantage of the **Waterfall** methodology. **[3]** *(AO1)*

7. *(from 1.2.2)* Explain the difference between **static** and **dynamic** linking. **[2]** *(AO1)*

8. *(from 1.1.1)* State the purpose of the **Accumulator (ACC)** register. **[1]** *(AO1)*